Universal gravitation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict how gravitational force changes with mass and separation.

SUCCESS CRITERIA - I can...

1. Use center-to-center separation.
2. Apply direct mass and inverse-square distance comparisons.
3. Identify equal forces on the two interacting bodies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gravity” and “separation” to explain your model or result. Sentence starters are optional.

gravity: an attractive interaction between masses

separation: distance between the specified objects or centers

inverse square: proportional to one divided by distance squared

01 Bring back an earlier idea

A rover travels around a bend at a steady 5 m/s.
Describe an observation that shows velocity changes even though the speed reading does not.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $F = Gm_1m_2/r^2$.
r is center-to-center separation. At fixed masses, doubling r makes the attractive force one quarter as large.

Worked reasoning: If separation doubles while both masses stay fixed, gravitational force becomes $1/2^2=1/4$ as large. A 20 N interaction would become 5 N.

Your turn: Sketch the two separations using centers.
Draw equal and opposite force arrows for each pair.

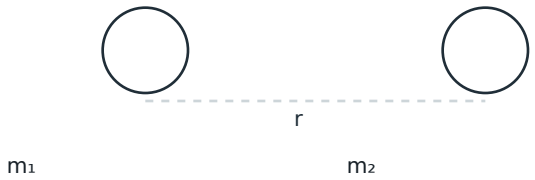
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Look and explain

Two equal spheres are shown.
Mark the distance r used by the gravity equation and draw a force on each sphere.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Build the solution

The original force is 12 N.
Predict it when one mass doubles, then separately when distance doubles.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

05 Use the idea

Does a large planet exert more force on a small moon than the moon exerts on the planet?
Compare forces and accelerations.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A runner pushes the ground backward with 60 N.
Name the partner force, its size and the object it acts on.
Explain its role in the runner's motion.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

07 Exit check

At separation r the force is 8 N. At 2 r with masses unchanged, what is it?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Starting from a force of 12 N, double one mass and double the separation.
Find the new force and explain both effects separately.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3